

The Voigt Effect for a Polarized Dense Alkali Vapor in a Small Magnetic Field



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POLARIZED HELIUM-3 AT WILLIAM AND MARY

Hyperpolarized noble gases using hybrid SEOP

Motivations in Experimental Nuclear Physics

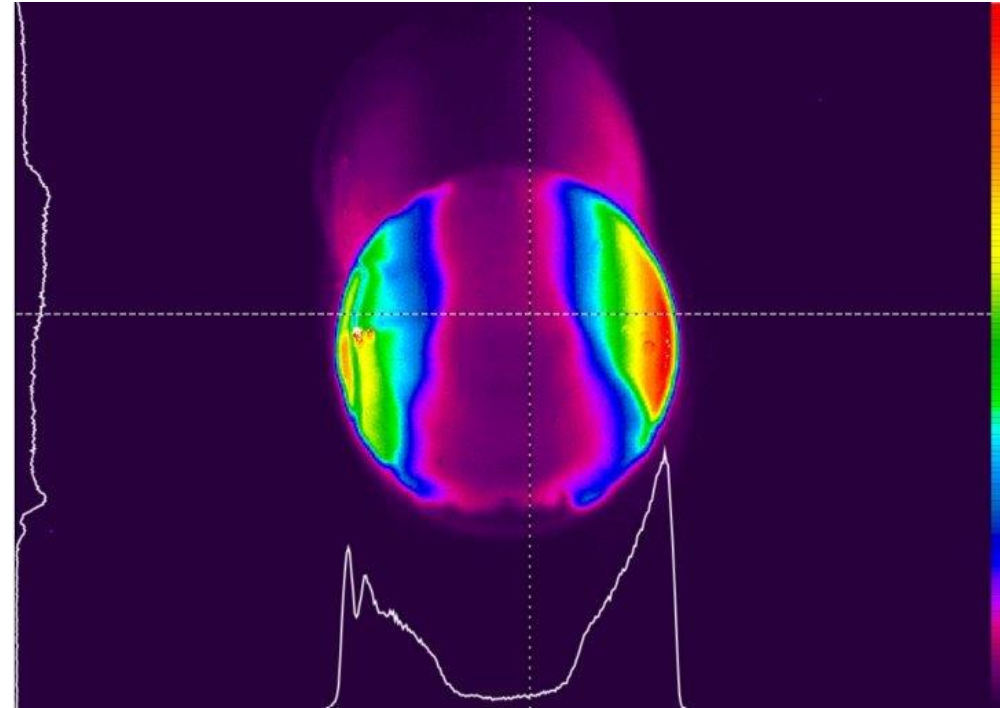
- Hyperpolarized ^3He targets used to study the spin structure of nucleons
- Targets polarized by first polarizing alkali, such as Rb and K, which transfers spin to ^3He nucleus
- Alkali is polarized via spin exchange optical pumping
- Spin/Alkali Polarization : The difference in proportions of alkali atoms with positive and negative spin



$$\mathcal{P} = \frac{N_+ - N_-}{N_+ + N_-}$$

Non-Uniform Polarization within a Cell

- Alkali polarization initially thought to be 100%
- $\mathcal{P}_{\text{He}}$ max is 70%
- Additionally, polarization may vary along the direction of pumping.
- Polarization can be measured through magneto-optical effects.



Spatial distribution of alkali polarization in a cell optically pumped from both ends [1]

$$\mathcal{P}_{He} = \langle \mathcal{P}_A \rangle (1 - e^{-t/\tau})$$

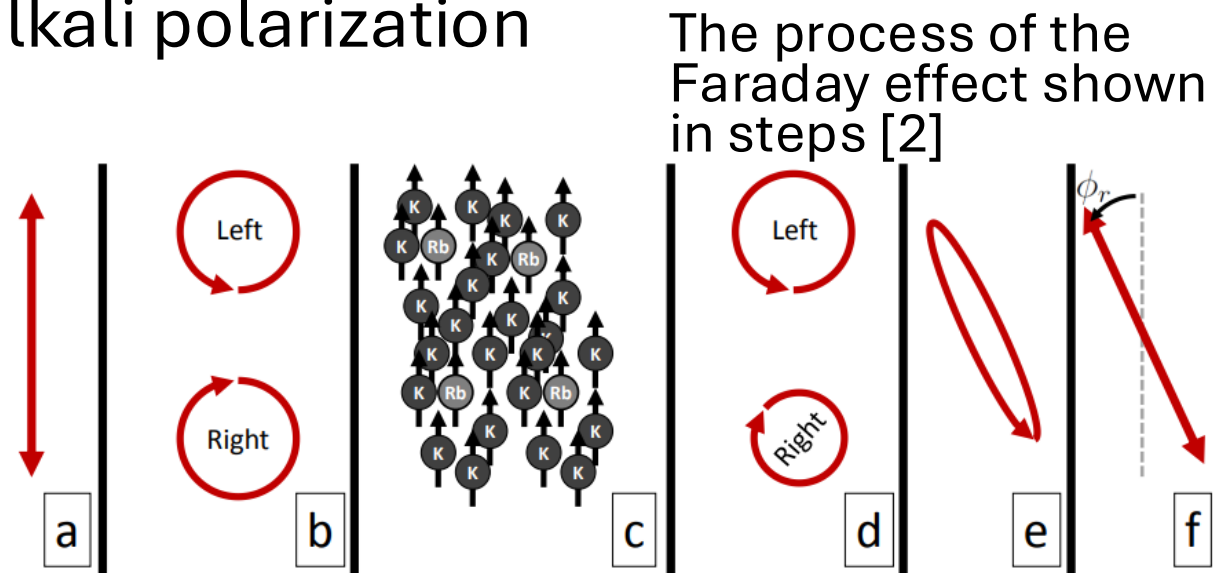
Polarization of the helium approaches the alkali polarization.

Faraday Effect in a Polarized Alkali

- Faraday effect occurs when light propagates in material parallel to a magnetic field
- Circular Birefringence - left and right circular components have difference indices of refraction
- Light's plane of polarization, is rotated by ϕ_F and intensity is attenuated
- These shifts in angle are proportional to the alkali polarization.
- Due to this, can be used to measure alkali polarization

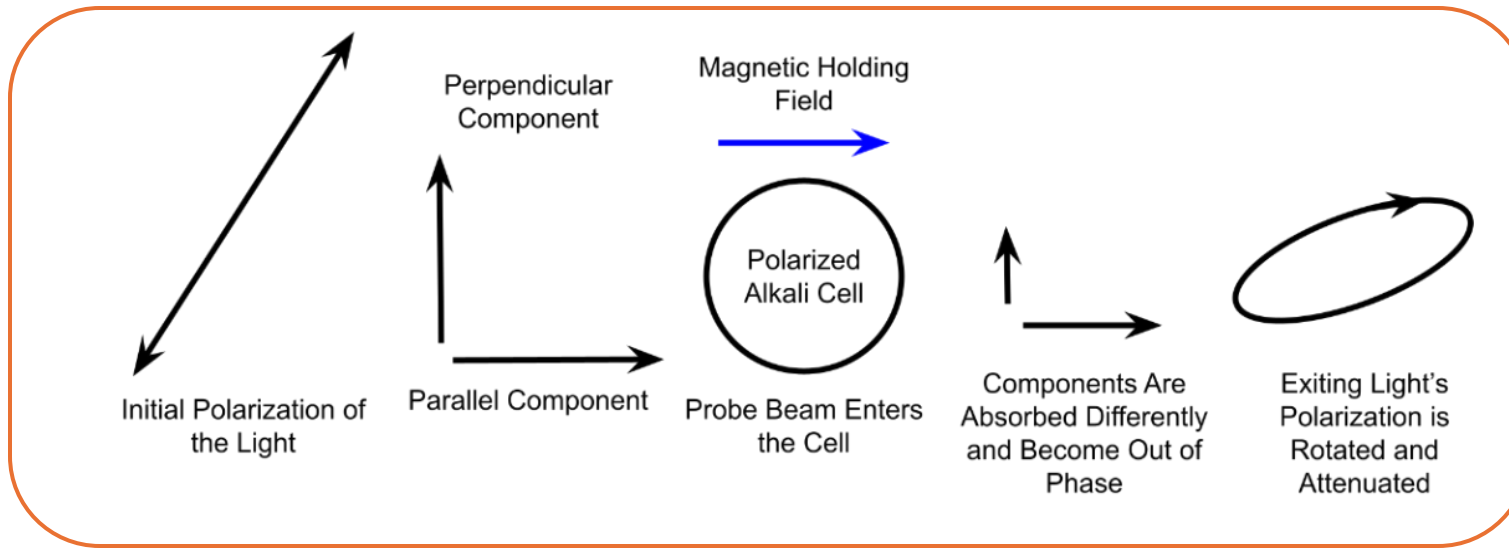
Faraday rotation in terms of alkali polarization [1]

$$\phi_F = -\frac{el[A]}{12mc\epsilon_0} f_F(\omega) \mathcal{P}$$

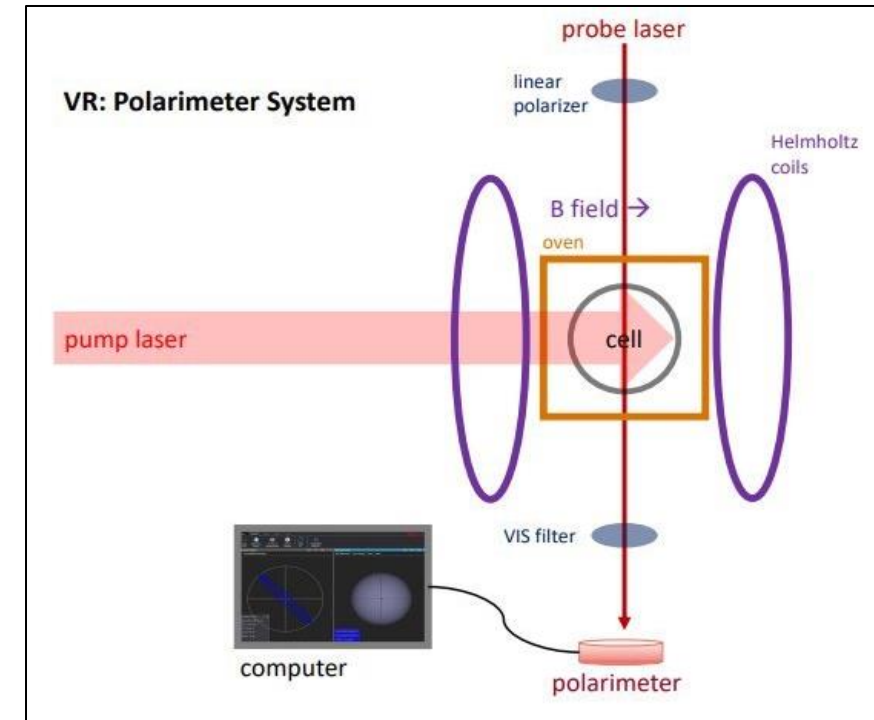


The Voigt Effect

- Occurs when light moves perpendicular to a magnetic field containing a magnetized material
- Linear birefringence – parallel and perpendicular components have different indices of refraction
- Could potentially be used to map the polarization of alkali within a cell



Experimental Setup [1]



Process for the Voigt Effect

Dipole Interaction and Polarizability

- Probe beam causes electric dipole transitions between S and P orbitals in the atoms
- Using time dependent perturbation theory, the expected dipole moment can be found.
- Dipole moment used to determine permittivity tensor of the gas
- Solving Maxwell's equations with the permittivity tensor reveals two eigenmodes for light to propagate with – source of linear birefringence

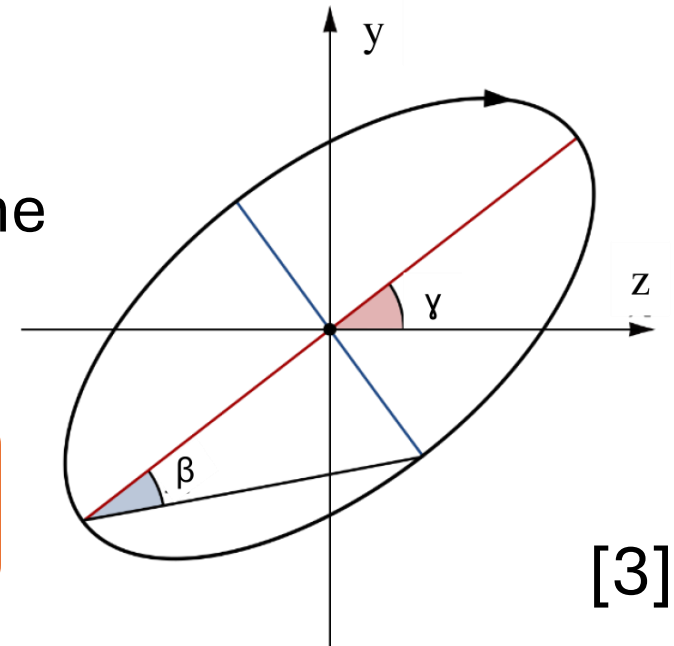
$$\overleftrightarrow{\varepsilon} = \begin{pmatrix} \varepsilon & -i[A]\delta/2 & 0 \\ i[A]\delta/2 & \varepsilon & 0 \\ 0 & 0 & \varepsilon \end{pmatrix}$$

$$k_{\parallel}^2 = \omega^2 \mu_0 \varepsilon$$
$$k_{\perp}^2 = \omega^2 \mu_0 \frac{(\varepsilon + [A]\delta/2)(\varepsilon - [A]\delta/2)}{\varepsilon}$$

Polarization Ellipse Under Linear Birefringence

- Both wavenumbers are complex, and their difference is proportional to P^2
- Polarization of light fully described by 4 Stokes parameters (I, Q, U, V)
- Assuming the probe beam to initially be linearly polarized with angle γ , Stokes parameters can be calculated
- From these parameters, the orientation and ellipticity of the polarization ellipse can be found.

$$\vec{D}(x, t) = D_0 e^{-i\omega t} (\cos(\gamma) e^{ik_{\parallel} x} \hat{z} + \sin(\gamma) e^{ik_{\perp} x} \hat{y})$$



Formulas for Voigt Rotation & Ellipticity Angles

- Both rotation, ϕ , and ellipticity, β , are proportional to the length travelled through the medium and the square of the alkali density.
- Both equations go to zero when not near a resonance
- Achieve maxima when probe beam is initially polarized at 45° degrees
- Most importantly, both are proportional to the square of alkali polarization.

Length Travelled: ℓ Initial Polarization Angle: γ
Alkali Density: $[A]$ Alkali Polarization: $\mathcal{P}$
Resonance Function: $f(\omega)$

$$\phi = \frac{e^4 \ell [A]^2}{24 m c \epsilon_0} \Im[f(\omega)] \sin(2\gamma) \mathcal{P}^2$$
$$\beta = \frac{e^4 \ell [A]^2}{24 m c \epsilon_0} \Re[f(\omega)] \sin(2\gamma) \mathcal{P}^2$$

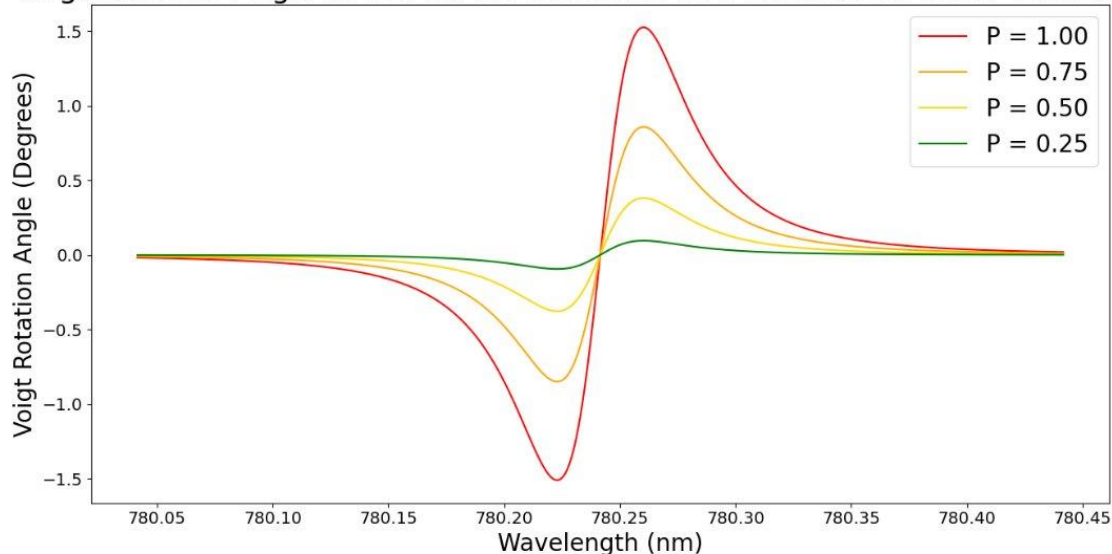
Dependence on Frequency

- Rotation and ellipticity increase as the frequency nears a D1 or D2 transition frequency.
- Rotation proportional to imaginary part of resonance function, $f(\omega)$ ellipticity related to real part

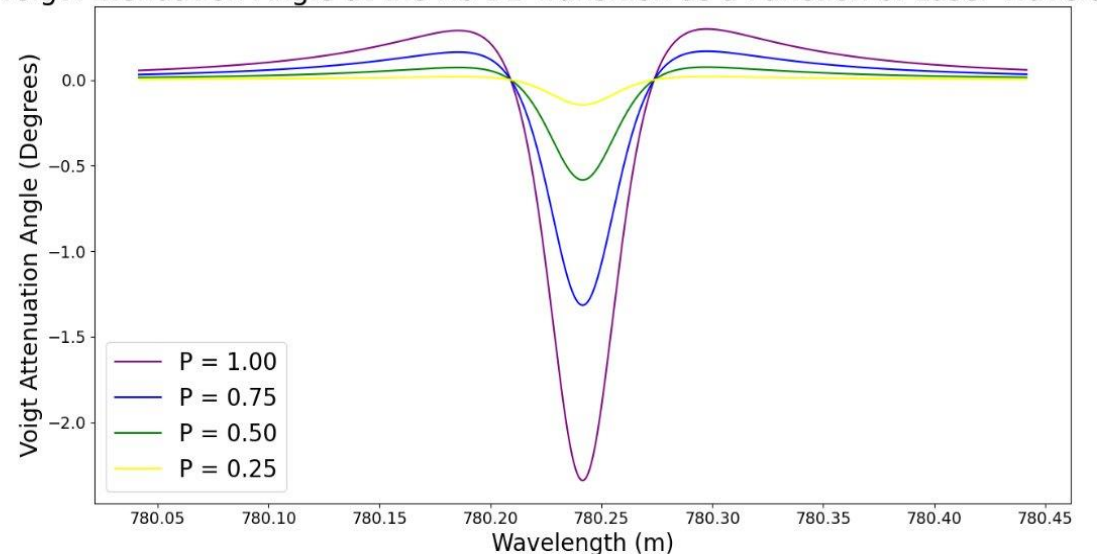
$$f(\omega) := \omega \left[\frac{(\mathcal{D}_2(\omega) - \mathcal{D}_1(\omega))^2}{6m\epsilon_0 + e^2[A](\mathcal{D}_1(\omega) + 2\mathcal{D}_2(\omega))} \right]$$

$$\mathcal{D}_N(\omega) = \frac{1}{\omega_{DN}} \frac{1}{\omega_{DN} - \omega - i\gamma_{DN}}$$

Voigt Rotation Angle at the Rb D2 Transition as a Function of Laser Wavelength

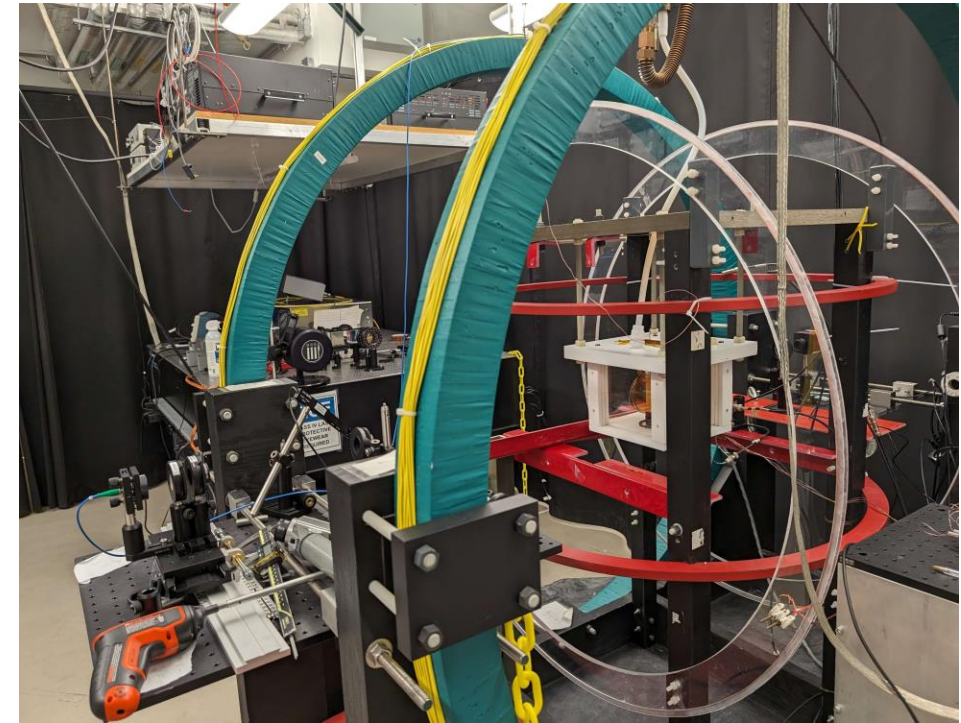


Voigt Attenuation Angle at the Rb D2 Transition as a Function of Laser Wavelength



Further Plans

- Begin attempting to measure alkali cell polarizations by measuring Voigt rotation and ellipticity
- More precise measurement needed since rotation ~ 1 degree
- Potential difficulty – probe beam may refract, and Faraday effect may skew results



Sources

- [1] L. Vannell, “Spatial Variability of Alkali-metal Polarization,” Bachelor’s Thesis, William & Mary, May 2023.
- [2] P. A. M. Dolph, “High-Performance Nuclear-Polarized ^3He Targets for Electron Scattering Based on Spin-Exchange Optical Pumping,” PhD thesis, University of Virginia, December 2010.
- [3]https://en.wikipedia.org/wiki/Stokes_parameters#/media/File:Polarisation_ellipse2.svg